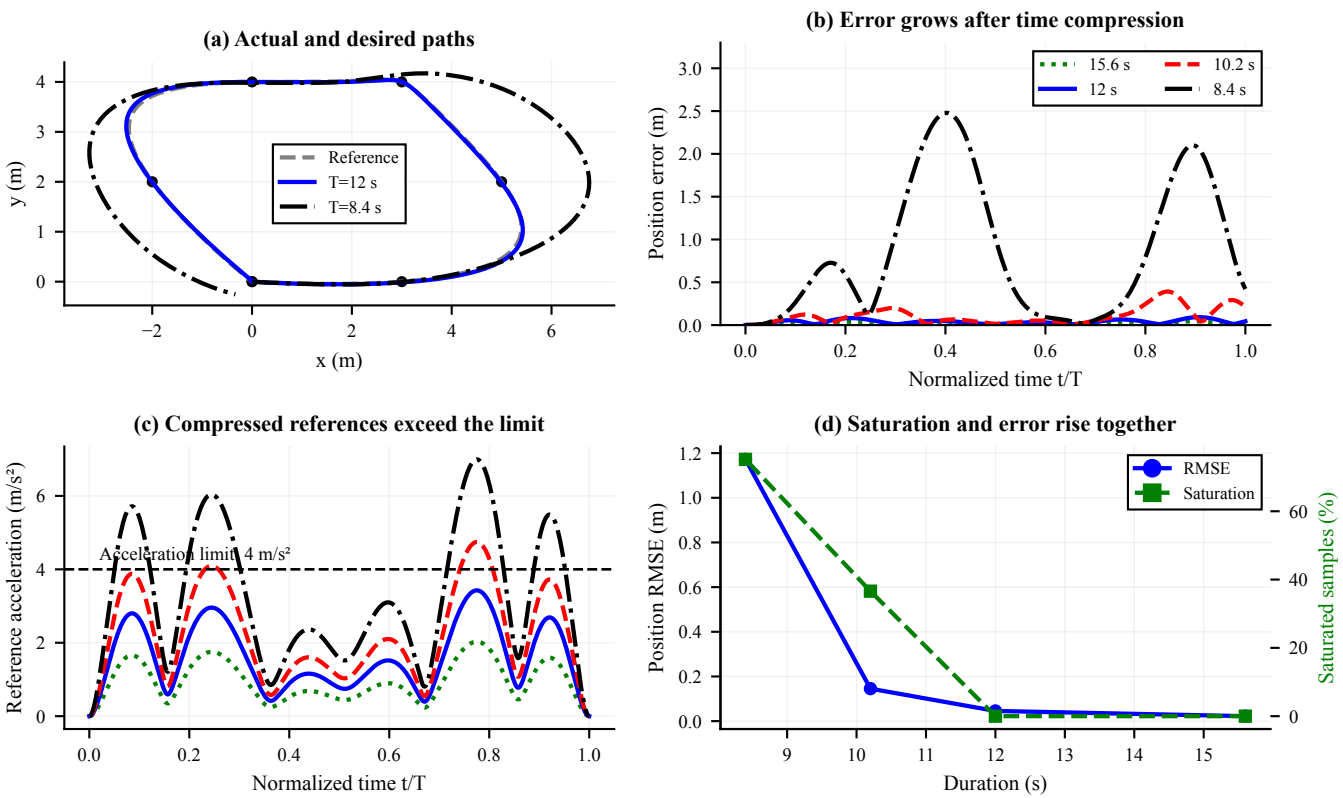

(a) Actual and desired paths
Reference
T=12 s
T=8.4 s
y (m)
x (m)
(b) Error grows after time compression
15.6 s
10.2 s
12 s
8.4 s
Position error (m)
Normalized time t/T
(c) Compressed references exceed the limit
Acceleration limit: 4 m/s²
Reference acceleration (m/s²)
Normalized time t/T
(d) Saturation and error rise together
RMSE
Saturation
Position RMSE (m)
Saturated samples (%)
Duration (s)